Hurt People Hurt People

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Abstract

Adverse childhood experiences are strongly associated with lifelong health, social, and economic outcomes, and increasing evidence suggests that their effects may propagate across generations. While empirical studies document the prevalence, transmission, and consequences of trauma, there is currently no unified framework linking these individual-level mechanisms to population-level dynamics. Here we combine a minimal analytical model with a data-informed simulation model to study the long-term evolution of trauma prevalence. The analytical model yields a closed-form threshold determining when trauma becomes self-sustaining, while the simulation model incorporates heterogeneous exposure levels and empirically grounded transmission patterns. The models show how demographic processes, intergenerational transmission, and recovery interact to produce stable endemic trauma levels or long-run decline, and establish the conditions under which interventions can eliminate endemic prevalence.

1 Introduction

Adverse childhood experiences (ACEs) are widely recognised as major determinants of lifelong health, social, and economic outcomes. The original ACE study demonstrated strong associations between early adversity and a range of adult disease risks [4]. Subsequent work has shown that exposure to multiple forms of adversity is common across populations [9]. Meta-analytic evidence further indicates that cumulative childhood adversity is strongly associated with both physical and mental health outcomes throughout the life course [7], with recent estimates suggesting that childhood maltreatment alone may account for a substantial fraction of global mental health burden [5].

Trauma may also propagate across generations. Intergenerational transmission of adversity has been documented across a range of social and developmental contexts [11, 16], and proposed mechanisms include behavioural pathways, socioeconomic constraints, and biological processes such as epigenetic regulation [22]. Developmental research further emphasises that early adversity can shape long-term stress responses and behavioural trajectories [17], while resilience and recovery processes can mitigate these effects under favourable conditions [12, 18].

Exposure to harsh or unpredictable environments has been linked to accelerated life-history strategies and earlier reproductive timing [3], and empirical studies have found associations between childhood adversity and fertility patterns [21]. Population prevalence is therefore shaped by the interplay between transmission, demographic processes, and recovery.

No theoretical framework yet links these mechanisms to the long-term population dynamics of trauma. Existing studies focus either on individual outcomes or on statistical correlations across cohorts. The first group documents associations between ACE exposure and adult health risk [4], synthesises dose–response relationships [7], and characterises developmental consequences of early toxic stress [17]. The second reports ACE prevalence estimates [9] and intergenerational transmission rates [11]. Neither approach addresses how trauma prevalence evolves over long time horizons or responds to structural interventions. Population modelling has proven useful for linking demographic processes to long-run outcomes [2], but has not been applied to trauma dynamics.

We develop a minimal analytical model that captures the essential mechanisms required for trauma to persist or decline in a population. This model isolates the roles of differential reproductive outcomes, intergenerational transmission, and healing, and yields explicit threshold conditions for self-sustaining trauma prevalence. We complement this with a data-informed simulation model incorporating heterogeneous exposure levels and empirically grounded transmission probabilities, which we use to explore how realistic distributions of adversity shape long-run population outcomes.

Throughout this work *adversity state* refers to an individual's level of cumulative ACE exposure, not to any single event. Within a generation, individuals move between lower and higher adversity states through exposure or healing; across generations, adversity propagates via behavioural, socioeconomic, and biological pathways. These mechanisms are illustrated in Figure 1.

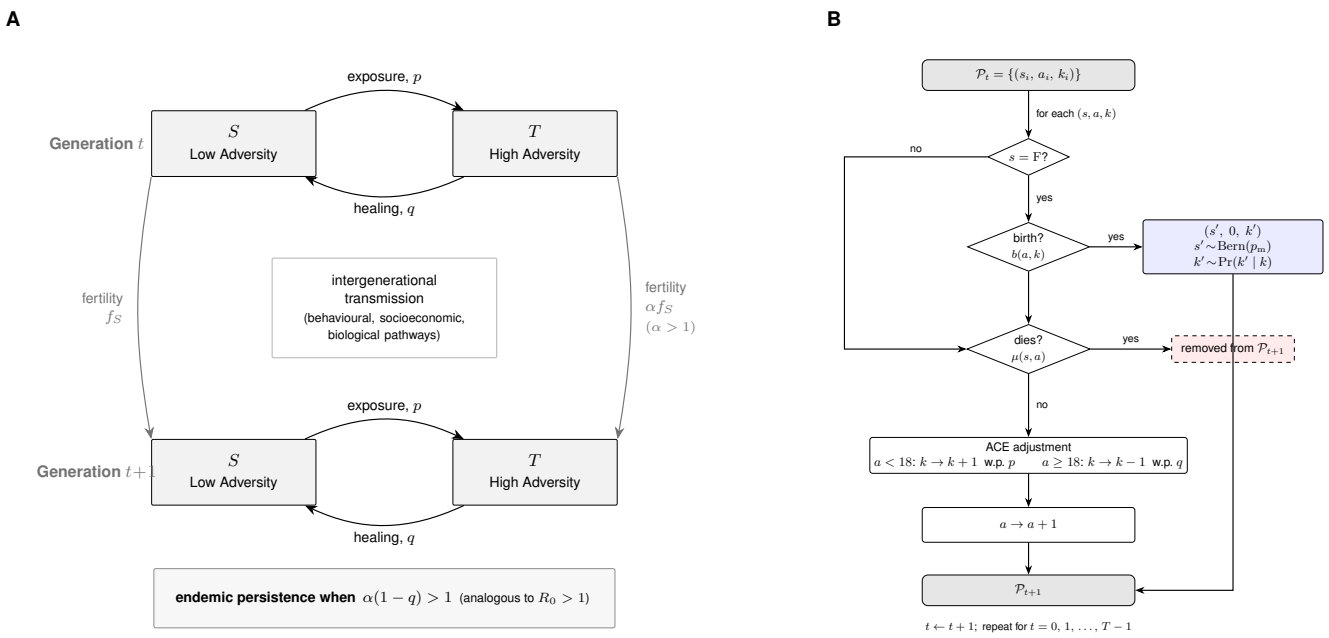


Figure 1: **Core model structure and simulation algorithm. A**, Individuals occupy one of two adversity states, S (lower adversity) or T (higher adversity), shown for two successive generations. Within each generation, individuals transition from S to T with probability p (exposure) and from T to S with probability q (healing). Across generations, T-state individuals reproduce at rate αf_S relative to the baseline fertility f_S of S-state individuals ($\alpha > 1$), reflecting empirically observed associations between adversity and accelerated life-history strategies. Adversity propagates intergenerationally via behavioural, socioeconomic, and biological pathways. The product $\alpha(1-q)$ acts as a demographic analogue of R_0: endemic trauma persists whenever $\alpha(1-q) > 1$. **B**, Flowchart of the individual-based simulation. At each annual step the three processes act sequentially on every member of the population denoted as $\mathcal{P}_t$. *Reproduction*, resolved first: females may give birth with probability $b(a, k)$, which combines a tempo effect, rescaling the baseline schedule about the youngest fertile age by the life-history pace $\pi(\bar{k}) = 1 + \gamma\bar{k}$ via effective age $\tilde{a} = 15 + (a - 15)\pi(\bar{k})$, and a quantum effect, scaling the rescaled probability by α for individuals with $k \geq 1$ ACEs and normalising by the population mean multiplier. Each newborn is assigned a sex s' drawn from a Bernoulli with parameter p_m and an ACE count k' drawn from the empirical intergenerational conditional distribution $\Pr(k' \mid k)$. *Mortality*, resolved second: each individual dies independently with the age- and sex-specific probability $\mu(s, a)$ taken from UN life tables; those who die are excluded from $\mathcal{P}_{t+1}$. *ACE adjustment*, applied to survivors: a child ($a < 18$) acquires one additional ACE with probability p (trauma exposure); an adult ($a \geq 18$) loses one ACE with probability q (healing). All survivors (with age incremented by one and ACE count updated) and all newborns are collected into $\mathcal{P}_{t+1}$.

2 Methods

2.1 Analytical model

Individuals are assumed to be either traumatised or not, so the population state is $x = (x_T, x_S)$, where x_T denotes the proportion of traumatised individuals and $x_S = 1 - x_T$ the remainder.

Let f_T and f_S denote the fitness of the respective subpopulations. Consistent with evidence linking adversity to accelerated life-history strategies and elevated fertility, we set $f_T = \alpha f_S$ with $\alpha > 1$ [3, 21].

We allow both the introduction of new trauma and the possibility of recovery [18, 22], modelled through two parameters:

- p: probability that a non-traumatised individual becomes traumatised,
- q: probability that a traumatised individual recovers.

Together, α, p, and q determine the balance between transmission, recovery, and differential reproduction, and hence whether trauma prevalence grows, declines, or stabilises over time (Figure 2A).

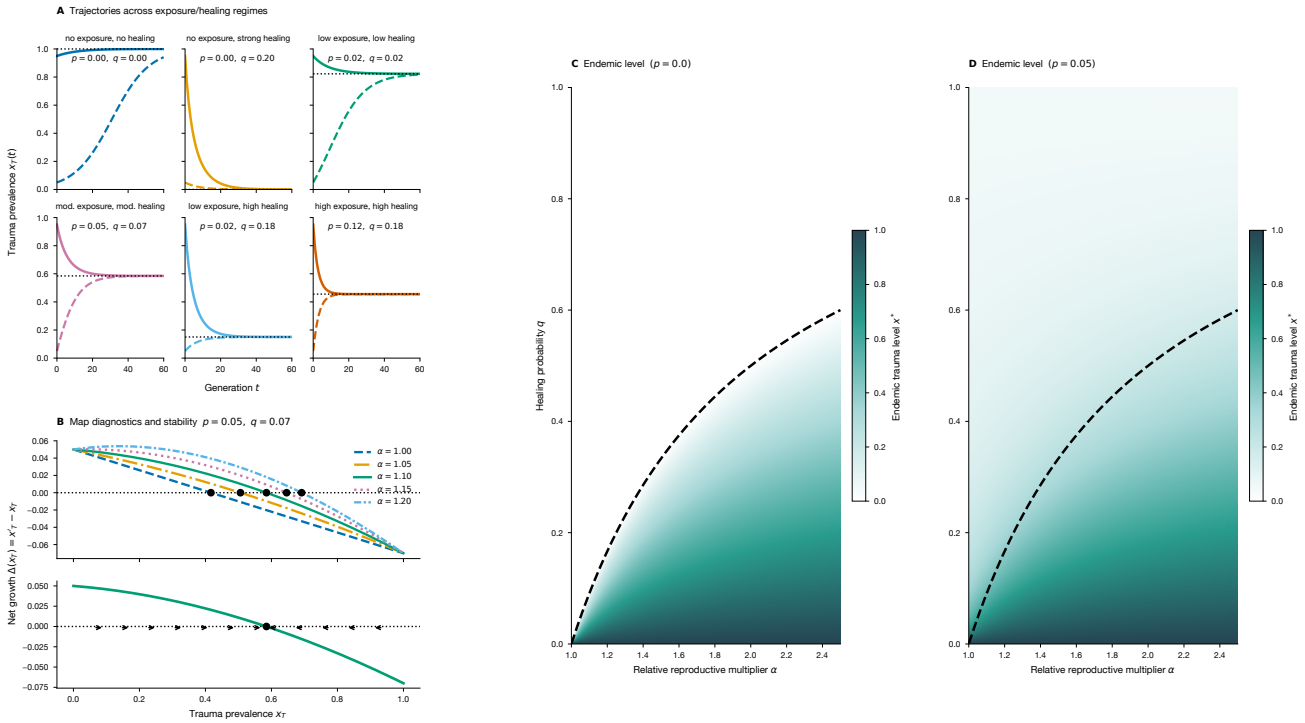


Figure 2: **Dynamics of trauma prevalence across mechanisms and parameters.** **A**, Time evolution of the traumatised proportion $x_T(t)$ under varying exposure (p) and healing (q) regimes. Trajectories from high and low initial conditions converge to the equilibrium level x^* (dotted line), illustrating how persistent exposure or recovery processes determine long-run prevalence. **B**, Analytical structure of the update map. Top: net growth function $\Delta(x_T) = x'_T - x_T$ for several values of the relative reproductive multiplier α, showing how equilibrium number and stability change with demographic amplification. Bottom: phase portrait for a representative parameter set, with arrows indicating the direction of population change. **C**, Parameter landscape for the invasion regime ($p = 0$). The coloured surface shows the endemic equilibrium level x^* as a function of α and healing probability q. The curve $q = 1 - \frac{1}{\alpha}$ separates parameter combinations for which trauma necessarily declines from those where it can persist endemically. **D**, Corresponding landscape with positive background exposure ($p = 0.05$). Introducing a non-zero exposure rate eliminates the trauma-free state, raising x^* across the entire parameter space and shifting the effective persistence boundary upward.

The population evolves according to the discrete replicator–mutation dynamic [8, 14],

$$x'_i = \sum_{j \in \{T,S\}} \frac{x_j f_j Q_{ji}}{\bar{f}}, \qquad i \in \{T, S\},$$

with mutation matrix

$$Q = \begin{pmatrix} 1-q & q \\ p & 1-p \end{pmatrix}.$$

The resulting update rule for the traumatised proportion is

$$x_T' = \frac{x_T f_T (1-q) + x_S f_S p}{\bar{f}}.$$

To assess persistence, consider the growth factor $\kappa_T = x_T'/x_T$. Substituting $f_T = \alpha f_S$ and $x_S = 1 - x_T$ yields

$$\kappa_T = \frac{1}{x_T} \frac{x_T\big(\alpha(1-q) - p\big) + p}{x_T(\alpha - 1) + 1}.$$

Values $\kappa_T > 1$ imply growth of the traumatised subpopulation, while $\kappa_T < 1$ imply decline, reflecting the standard growth criterion used in evolutionary dynamics [6, 14].

Interior equilibria satisfy $x_T' = x_T$, leading to the quadratic

$$(\alpha - 1)x_T^2 + (1 + p - \alpha + \alpha q)x_T - p = 0,$$

Because the constant term equals $-p < 0$ while evaluation at $x_T = 1$ gives $\alpha q \geq 0$, continuity guarantees a root in $(0, 1)$ whenever $p, q > 0$. This root defines the endemic level x_T^*, showing that any persistent introduction of trauma produces a stable long-run prevalence (Figure 2A).

When no new trauma is introduced ($p = 0$), persistence depends on invasion from rarity. Taking the limit $x_T \to 0$ gives

$$\lim_{x_T \to 0} \kappa_T = \alpha(1-q),$$

so trauma spreads from low prevalence precisely when

$$\alpha(1-q) > 1 \qquad \Longleftrightarrow \qquad q < 1 - \frac{1}{\alpha}.$$

This condition acts as a demographic threshold separating extinction from endemic persistence, shown in Figure 2C.

Long-run trauma prevalence is therefore governed by the balance between transmission, recovery, and demographic reinforcement. A sufficient reproductive advantage can sustain trauma across generations even without continual exposure, and recovery must cross a critical threshold to eliminate it. The system behaves, in this respect, like an infectious agent with a demographic amplification mechanism.

Figure 2B–C summarises the equilibrium structure. The next section embeds these mechanisms in a data-informed simulation that adds demographic stochasticity, age structure, and empirically grounded parameterisation.

3 Individual-based simulation model

To explore how stochasticity, finite-population effects, and demographic heterogeneity modify the analytical predictions, we built an individual-based simulation of the same mechanisms.

Individual state and population dynamics

Each individual is characterised by $(s, a, k, \bar{k})$ denoting sex $s \in \{\mathrm{M}, \mathrm{F}\}$, integer age $a \geq 0$, current ACE count $k \in \{0, 1, \ldots, 8\}$, and the count accumulated by the end of childhood $\bar{k} \geq k$, which is fixed from age 18 onwards. The population at time t is the multiset $\mathcal{P}_t = \{(s_i, a_i, k_i, \bar{k}_i)\}_{i=1}^{N_t}$, where N_t is the (stochastic) population size. The model evolves in discrete yearly steps; in each step three processes act sequentially on every individual: reproduction, mortality, and within-life ACE adjustment. Reproduction precedes mortality, so an individual who dies during year t may still give birth in that year.

Mortality

Each individual dies in year t with probability $\mu(s, a)$, drawn from age- and sex-specific life-table values based on UN World Population Prospects data [20]. Mortality is assumed independent of ACE count, so demographic selection operates exclusively through fertility.

Fertility and ACE transmission

Reproduction is restricted to females and is resolved before mortality, so a female who dies during the year may still give birth in it. The baseline probability of a birth in a given year for a female of age a is $b_0(a)$, taken from UN age-specific fertility schedules [19]. Two empirically motivated mechanisms modify this baseline for ACE-exposed individuals.

Tempo effect (earlier reproductive timing)

Consistent with evidence that adversity is associated with accelerated life-history strategies and earlier age at first birth [21], the fertility schedule is shifted earlier in proportion to ACE exposure.

We distinguish two counts. Write k for an individual's current ACE count, which healing reduces, and $\bar{k}$ for the count accumulated by the end of childhood. Since healing acts only on adults, $\bar{k}$ is the total an individual was ever exposed to and is fixed from age 18 onwards, while $k \leq \bar{k}$ is what remains after any recovery. The tempo effect reads $\bar{k}$ and the quantum effect reads k.

A female then evaluates her birth probability at an effective age

$$\tilde{a} = a_0 + (a - a_0)\,\pi(\bar{k}), \qquad \pi(\bar{k}) = 1 + \gamma\bar{k},$$

where $a_0 = 15$ is the youngest age in the fertility schedule and $\pi(\bar{k})$ is the *life-history pace*. The birth probability is scaled by the same factor, $b(a,k) \propto \pi(\bar{k})\, b_0(\tilde{a})$, so that the schedule is compressed rather than translated and completed fertility is preserved.

We use $\bar{k}$ rather than k here because life-history pace is calibrated during development [3] and is not reset by later recovery. Reading the current count instead would allow an individual who had aged out of her compressed fertile window to re-enter it by healing, so that recovery manufactured fertility rather than retiming it. We set $\gamma = 0.314$, calibrated so that a single ACE lowers the mean age at childbearing by three years, matching the estimate of Williams and Cheadle [21]. We rescale rather than translate because a rigid shift of an age-bounded schedule moves the fertile window below a_0, and for large k pushes it off the schedule entirely, which destroys completed fertility and produces implausible ages at birth. Under rescaling the shift in mean age at childbearing saturates on its own, from three years at $\bar{k} = 1$ to nine years at $\bar{k} = 8$, so no cap is required. This implements a purely timing adjustment: the expected number of births over a full reproductive span is unchanged to within 0.2% across $\bar{k}$. Net reproduction, which weights each birth by the probability of surviving to that age, rises slightly (2.1% at $\bar{k} = 8$) because earlier childbearing is less discounted by mortality. This is a demographic consequence of the timing shift rather than an additional fertility mechanism, and its size depends on the mortality schedule. The resulting schedules are shown in Figure 3A, and their consequences for the timing and the total of reproduction in Figure 3B.

Quantum effect (differential reproductive intensity)

Conditional on effective age $\tilde{a}$, the birth probability of a female with $k \geq 1$ ACEs is scaled by a multiplicative factor $\alpha > 1$:

$$b(a,k) = \frac{\pi(\bar{k})\, b_0(\tilde{a})}{\bar{\alpha}} \times \begin{cases} 1 & \text{if } k = 0, \\ \alpha & \text{if } k \geq 1, \end{cases}$$

where $\bar{\alpha}$ is the population mean multiplier, weighted by the fertility each individual would have at $\alpha = 1$,

$$\bar{\alpha} = \frac{\sum_i \pi(\bar{k}_i)\, b_0(\tilde{a}_i)\, \alpha^{\mathbf{1}[k_i \geq 1]}}{\sum_i \pi(\bar{k}_i)\, b_0(\tilde{a}_i)},$$

summed over females. Dividing by $\bar{\alpha}$ holds the total expected number of births at its $\alpha = 1$ value, so that α redistributes reproduction towards exposed individuals rather than raising it for the population as a whole, while the ratio between the two groups remains exactly α (Figure 3C). This is the normalisation the replicator dynamic of Section 2 performs when it divides by mean fitness $\bar{f}$; without it, raising α would raise the population growth rate as well as the relative standing of the exposed, and the two models would not share a parameter. Under a coarse-grained binary classification (ACE-free versus ACE-exposed), this corresponds directly to the relative fitness parameter of the analytical model, in which $f_T = \alpha f_S$. The simulation therefore embeds the replicator structure of the analytical model within an age-structured, stochastic demographic framework.

Intergenerational ACE transmission

The ACE count of each newborn is drawn from an empirically estimated conditional distribution $\Pr(k_{\text{child}} \mid k_{\text{mother}})$, based on reported intergenerational ACE correlations [16]. Denoting the four maternal exposure categories as $\{0, 1, 2\text{–}3, \geq 4\}$, the conditional probabilities are summarised in Table 1. Offspring ACE counts in the range $\{2, 3\}$ and $\{4, \ldots, 8\}$ are distributed uniformly within their respective groups. This mechanism allows adversity to propagate across generations in the absence of ongoing environmental shocks, providing the simulation analogue of the transmission parameter p in the analytical model.

Within-life ACE dynamics

ACE counts may change within a lifetime through two complementary processes that correspond directly to the analytical parameters p and q.

Table 1: Conditional probabilities $\Pr(k_{\text{child}} \mid k_{\text{mother}})$ for offspring ACE category given maternal ACE category, expressed as percentages with 95% confidence intervals. Source: [16].

	Maternal ACE category			
Child ACE category	0	1	2–3	≥ 4
0	42.6 (38.4–46.7)	37.8 (32.4–43.1)	30.4 (24.7–36.1)	25.1 (16.2–34.0)
1	25.8 (21.9–29.6)	29.7 (23.9–35.5)	28.4 (22.2–34.7)	23.2 (14.1–32.2)
2–3	25.8 (21.9–29.7)	21.5 (16.3–26.7)	21.3 (15.7–26.8)	23.8 (15.6–32.0)
≥ 4	5.8 (4.0–7.7)	11.0 (6.3–15.6)	19.9 (13.7–26.1)	27.9 (18.9–36.9)

- *Trauma exposure* (p): a child aged $a < 18$ acquires one additional ACE with probability p per year, subject to the ceiling $k \leq 8$. This represents environmental and social transmission of adversity during the developmental window.
- *Healing* (q): an adult aged $a \geq 18$ reduces their ACE count by one with probability q per year, subject to the floor $k \geq 0$. This represents recovery processes including therapeutic intervention, social support, and self-forgiveness [13, 18].

The net annual change in ACE count for individual i is therefore

$$\Delta k_i = \begin{cases} +1 & \text{if } a_i < 18 \text{ and } U_i < p, \\ -1 & \text{if } a_i \geq 18 \text{ and } U_i < q, \\ 0 & \text{otherwise,} \end{cases}$$

where $U_i \sim \text{Uniform}(0, 1)$ is drawn independently each year. p and q play the same roles in the simulation as in the analytical model: p governs the rate of acquiring adversity and q the recovery rate. The two processes act on disjoint phases of life, so a child cannot heal while still accumulating adversity (Figure 3E).

Initial conditions

Simulations are initialised from a population pyramid based on UK Office for National Statistics census data [15], from which ages and sexes are sampled according to empirical proportions. Initial ACE counts are drawn from sex-stratified prevalence distributions reported in the Canadian Longitudinal Study on Aging [9].

Simulation protocol

We swept the three parameters that the analytical model shares with the simulation. The healing probability q takes 61 values from 0.00 to 0.60 in steps of 0.01; the reproductive multiplier α takes ten values from 1.05 to 1.50 in steps of 0.05; and the background exposure probability p takes the eleven values from 0.00 to 0.10 in steps of 0.01, together with 0.20 as a saturated case retained for the supplementary material. This gives 7,320 parameter combinations.

Each combination was simulated for 50 independent replicates of 200 years, starting from 10,000 individuals, for a total of 366,000 runs. Newborns are assigned male with probability 0.51, and an individual is counted as traumatised when $k \geq 1$. Summary quantities are averaged over the years from 100 onwards, discarding the first 100 years as a warm-up during which the population relaxes from the initial age structure and ACE distribution to its long-run state; Figure 5A–B shows that this is comfortably longer than the transient.

Every run is seeded explicitly, so the sweep reproduces exactly. Each parameter combination draws from its own block of 50 consecutive seeds, and the blocks are disjoint, so replicates are independent both within and across combinations. A single generator is shared between the construction of the initial population and the run itself, so each replicate consumes one continuous random stream. Between-replicate variation is shown throughout as an inter-quartile range or a standard deviation.

Relationship to the analytical model

The simulation preserves the core dynamical structure: α, p, and q enter with the same meanings as in the replicator–mutation model. It replaces the mean-field assumption with explicit demographic stochasticity and age structure, and the binary ACE state with a discrete count on $\{0, \ldots, 8\}$. In the limit of large population size and binary aggregation of the ACE count, the simulation dynamics converge to the analytical predictions; departures reflect stochasticity and heterogeneity.

To identify the boundary between high- and low-adversity parameter regimes in the simulation output we applied k-means clustering ($k = 2$) to the log-transformed mean ACE surface; full details of this procedure are given in Supplementary Information .5.

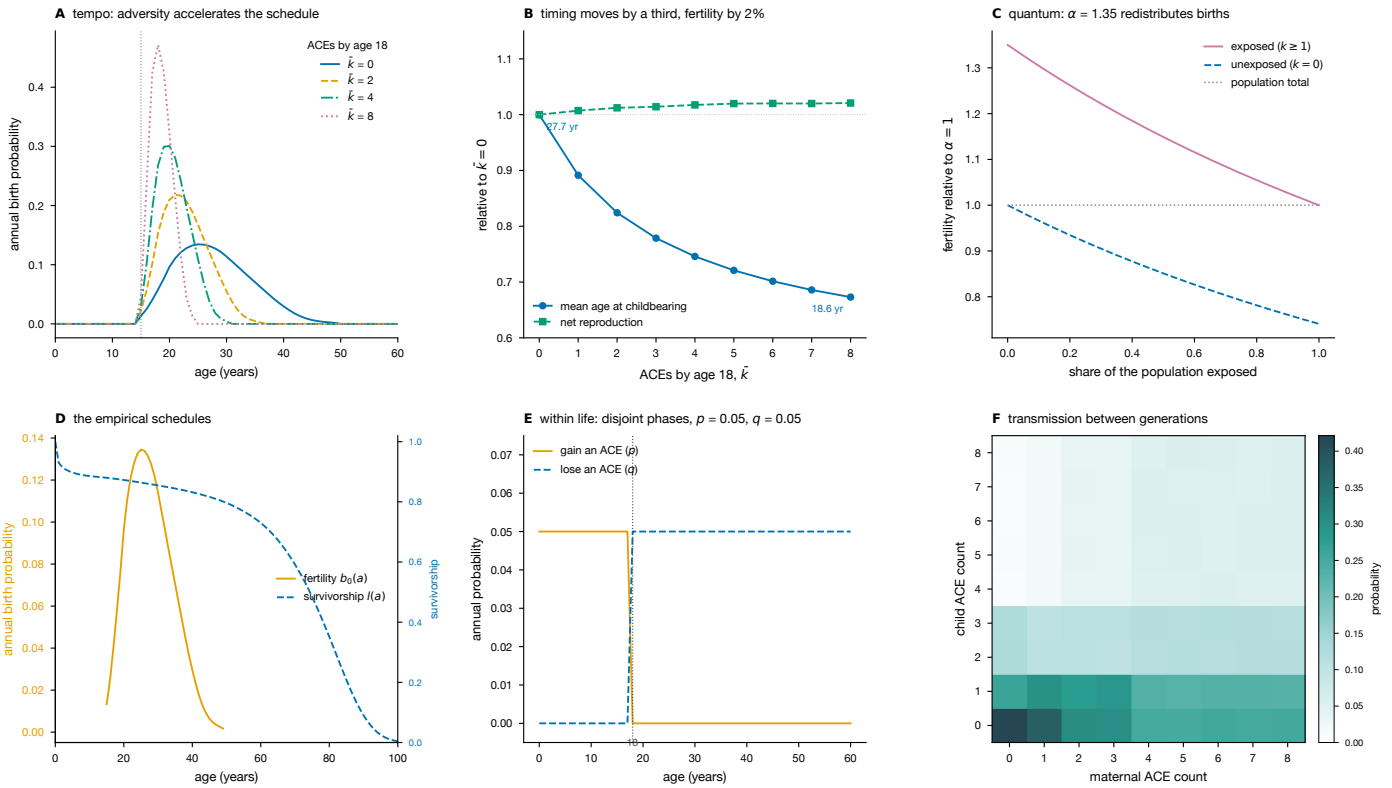


Figure 3: **Specification of the simulation model.** Three mechanisms determine how adversity moves through the population: its effect on fertility (**A**–**C**), its change within a lifetime (**E**), and its transmission between generations (**F**). **A**, Annual birth probability against age for four values of the completed ACE count $\bar{k}$. Adversity accelerates the life-history schedule, rescaling the age axis about the youngest fertile age a_0 by the pace $\pi(\bar{k})$ and scaling the probability by the same factor. The fertile window therefore starts at a_0 for every $\bar{k}$ and is compressed rather than moved, and the higher peaks are the pace factor. **B**, Consequences for the two components of reproduction, both shown relative to their value at $\bar{k} = 0$ so that a change in one can be read against a change in the other; separate axes would let the scaling decide which curve appears to move. The mean age at childbearing falls by a third, from 27.7 to 18.6 years (annotated), while net reproduction rises by 2.1%. That rise is not additional fertility: the expected number of births over a full reproductive span is unchanged to within 0.2%, and net reproduction weights each birth by the probability of surviving to that age, so bringing childbearing forward discounts it less by mortality. The tempo effect therefore changes when individuals reproduce rather than how much. **C**, The quantum effect. Fertility is normalised by the population mean multiplier, so that α redistributes reproduction towards exposed individuals while leaving the total unchanged (dotted line). The ratio between the two curves is α at every composition. This is the normalisation the replicator dynamic performs when it divides by mean fitness; without it α would raise the population growth rate rather than the relative standing of the exposed. **D**, The two empirical schedules the model is built on: the UN age-specific fertility schedule $b_0(a)$ [19] and the survivorship $l(a)$ implied by the UN life table [20]. **E**, Within-life dynamics. Exposure acts only during childhood and healing only during adulthood, so the two processes occupy disjoint phases and an individual cannot heal from adversity while still accumulating it. **F**, The intergenerational transmission matrix $\Pr(k' \mid k)$, sampled from the model itself. Mass shifts upwards as maternal exposure rises; the flat bands reflect the uniform allocation within the reported 2–3 and ≥ 4 categories of Table 1.

Figure 4 illustrates the baseline demographic behaviour of the simulation across six parameter combinations ($q \in \{0.00, 0.01, 0.05\}$, $\alpha \in \{1.05, 1.35\}$, no background exposure): population size and mean age settle within a few decades across all combinations (panels A–B), and the growth-corrected survivorship curves fall on the theoretical life-table predictions irrespective of ACE parameters, confirming that mortality is ACE-independent (panel C). The correction is needed because a population snapshot yields period rather than cohort survivorship; the caption describes it. Population age pyramids at 50-year intervals (panels D–H) show the transition from the initial UK age structure to the long-run steady-state shape for a representative parameter set ($q = 0.05$, $\alpha = 1.35$, no background exposure).

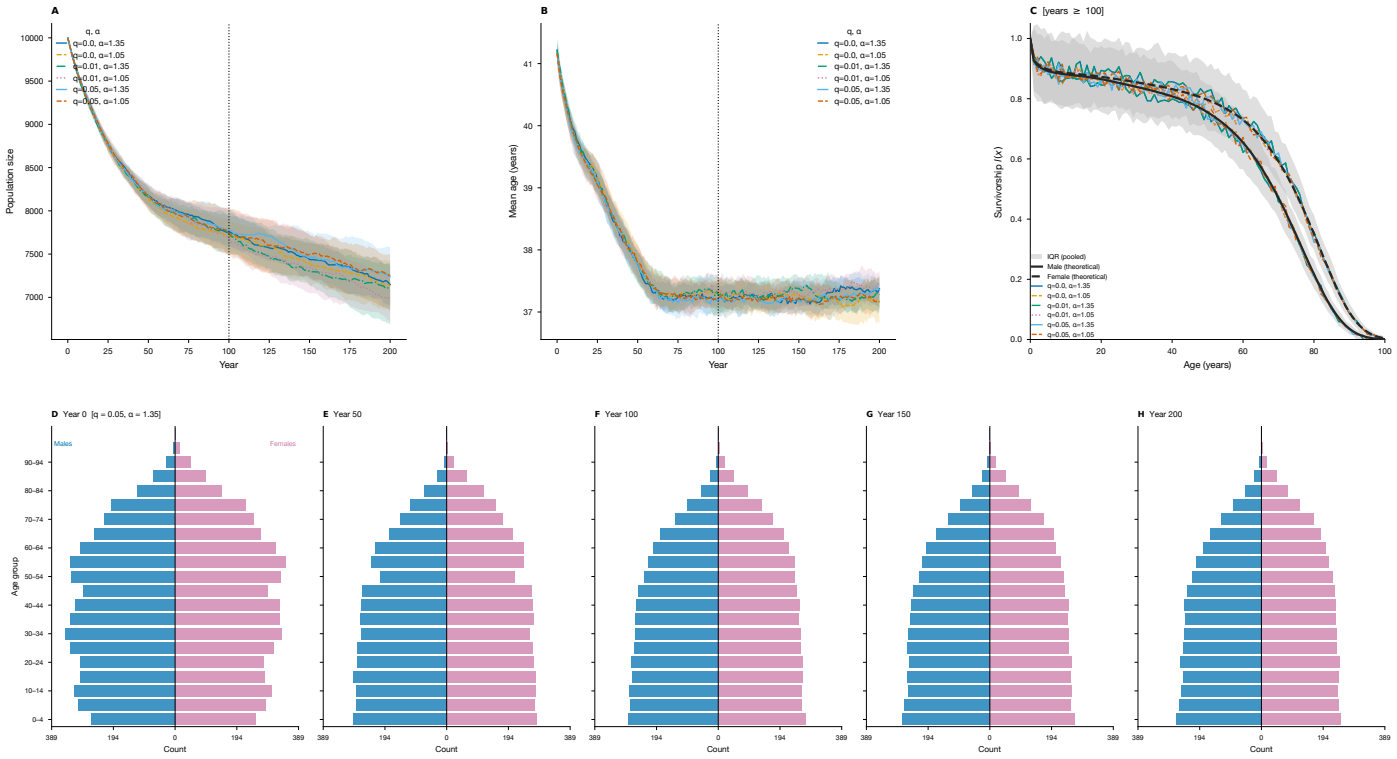


Figure 4: **Population-level demographic dynamics of the simulation.** Results shown for six parameter combinations ($q \in \{0.00, 0.01, 0.05\}$, $\alpha \in \{1.05, 1.35\}$, no background exposure) to demonstrate that the demographic dynamics are robust across the parameter space explored. **A**, Trajectories of total population size over 200 years (median ± IQR across replicates, one line per parameter combination). **B**, Mean age of the population over time, with the same summary. **C**, Survivorship $l(x)$ recovered from simulation snapshots after year 100. A snapshot gives the period ratio $N(x)/N(0)$, which equals the life-table $l(x)$ only in a stationary population: where the population grows at rate r the stable age distribution is proportional to $l(x)\mathrm{e}^{-rx}$, so the raw ratio understates survival by that factor. We therefore estimate r for each replicate by regressing log population size on time across the snapshot years, and rescale by e^{rx}. The correction is substantial, reaching a factor of 1.8 by age 80 at $r = 0.0073\,\mathrm{yr}^{-1}$. One pair of median lines (male and female sharing the same colour and line style) is shown per parameter combination, and the grey band shows the pooled inter-quartile range across all parameter combinations and replicates. Black lines show the theoretical survivorship curves derived directly from the UN life-table probabilities (solid: male; dashed: female). All six parameter combinations collapse onto those curves, separating only into the male and female pair the life table predicts. This confirms both that the simulated age structure is consistent with the embedded mortality schedule and that it is insensitive to the ACE transmission parameters, as expected given that mortality is modelled as ACE-independent. We note that a single scalar r per replicate corrects the whole age profile, so the agreement is not a matter of fitting: were the mortality schedule misapplied, no choice of r would recover the theoretical curves. The collapse is in addition evidence that the population has reached a stable age distribution by year 100, since the identity used to correct it holds only in that regime. **D**–**H**, Population age pyramids (median counts across replicates) at years 0, 50, 100, 150, and 200 for the representative parameter set ($q = 0.05$, $\alpha = 1.35$, no background exposure), showing the transition from the initial UK census age structure to the long-run steady-state shape imposed by the empirical fertility and mortality schedules.

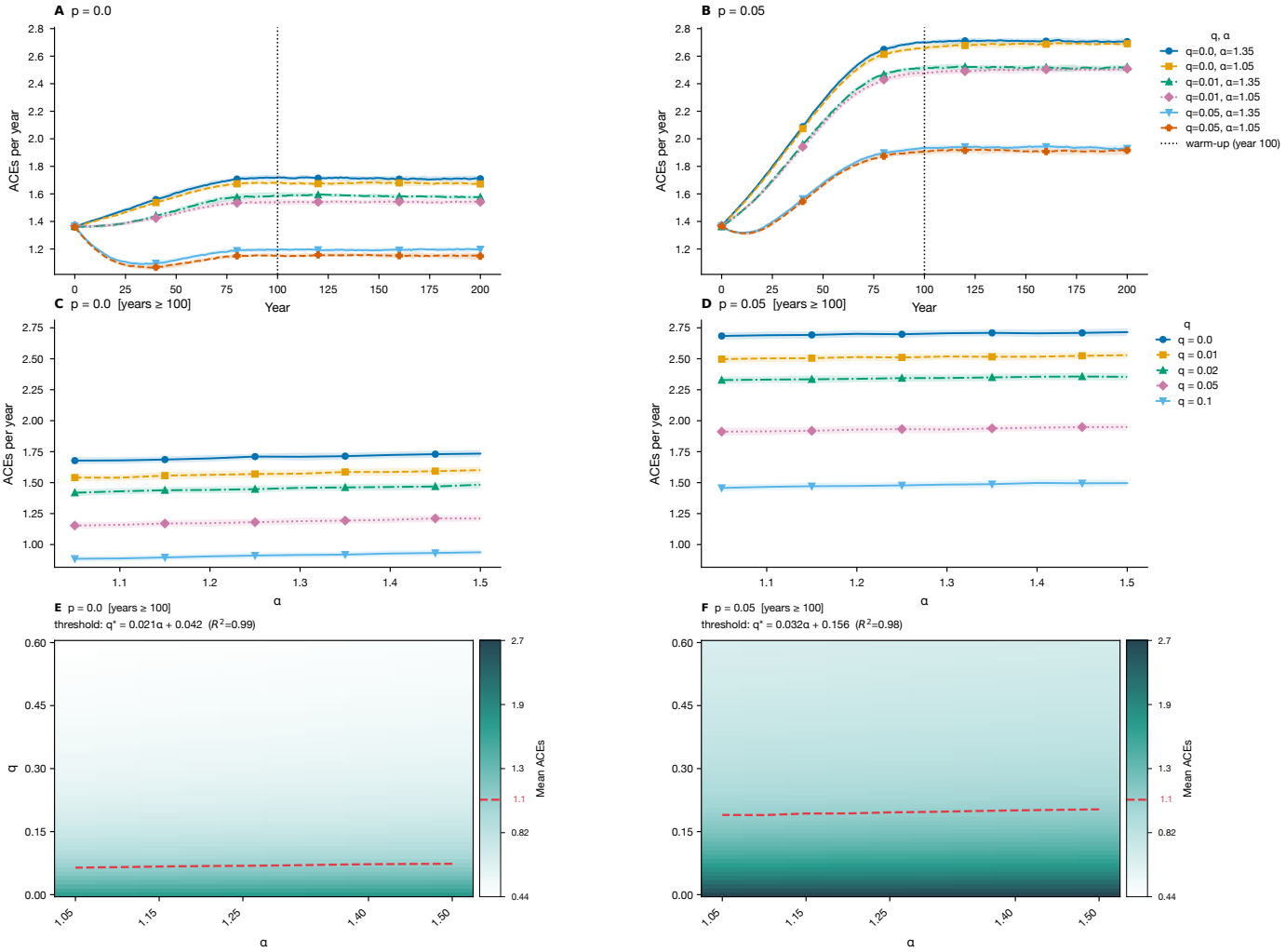


Figure 5: **Simulation outcomes across parameter regimes (mean ACE count).** Panels are arranged in three rows and two columns; the left column shows results without background trauma exposure ($p = 0$) and the right column with positive background exposure ($p = 0.05$). **A**–**B**, Time trajectories of mean ACEs per person for six combinations of healing probability q and reproductive multiplier α, shown as median ± IQR across simulation replicates. The dashed vertical line marks the end of the 100-year warm-up period. **C**–**D**, Mean ± SD of mean ACE count as a function of α, with one curve per healing probability. Higher α and lower q consistently raise long-run adversity levels, with the effect amplified under positive background exposure. **E**–**F**, Heatmaps of the time-averaged mean ACE count in the (q, α) parameter plane, shown on a $\log(1 + \cdot)$ scale. The dashed red contour marks the k-means threshold separating high- and low-adversity regimes, computed once on the surfaces pooled over both values of p (Supplementary Information .5). Each panel title gives the least-squares fit $q^* = m\alpha + c$ to that contour. Here q is an annual rate; on a per-generation scale (Supplementary Information .2) the boundary lies above the analytical threshold $q = 1 - \alpha^{-1}$ and is markedly flatter in α, as we discuss in Section 4. Corresponding results for the proportion-traumatised metric are shown in Figure 6.

4 Discussion

We have combined a minimal analytical model of trauma transmission with a data-informed individual-based simulation of the same mechanisms.

Our central result is a closed-form threshold, $q < 1 - 1/\alpha$, separating parameter regimes in which trauma necessarily declines from those in which it persists endemically. When both a positive background exposure and a positive recovery rate are present ($p, q > 0$), the quadratic fixed-point equation admits an interior root, so any sustained environmental adversity produces a stable long-run prevalence rather than eventual elimination. We show in Supplementary Information .1 that this root is unique and globally stable, and that the threshold is an exchange of stability between the trauma-free and endemic states rather than merely a sign condition.

The threshold condition $\alpha(1 - q) > 1$ is structurally analogous to the basic reproduction number R_0 in infectious-disease epidemiology [6], with the product $\alpha(1 - q)$ playing the role of R_0. Unlike standard SIS and SIR frameworks, this system contains a demographic amplification term: adversity is not merely transmitted but is reinforced by a reproductive advantage that standard epidemic models lack.

What the simulation adds

The two models do not share a time step, and comparing their thresholds requires the mapping we derive in Supplementary Information .2: the analytical parameters act once per generation on a binary state, while the simulated ones act once per year on a graded count and are confined to disjoint phases of life.

On that common footing the two thresholds differ in both level and slope, and the way they differ is itself a result. Converted to a per-generation probability, the fitted boundary without background exposure, $q^*(\alpha) = 0.021\alpha + 0.042$ ($R^2 = 0.99$), runs from 0.47 at $\alpha = 1.05$ to 0.52 at $\alpha = 1.50$. The analytical threshold over the same range runs from 0.05 to 0.33. The simulated boundary therefore sits above the analytical one throughout, by a factor of 1.6 at $\alpha = 1.5$ rising to an order of magnitude as $\alpha \to 1$, and it is markedly flatter: it moves by 10% across the swept range of α where the analytical threshold moves by a factor of seven.

The two discrepancies have different sources. The level reflects structure the analytical model omits. Healing reaches an individual only during the years between adulthood and reproduction, and it removes one adverse experience at a time rather than clearing the state outright, so suppressing adversity in an age-structured population is harder than a mean-field account implies. The flatness reflects something else. An individual has $D = 9.7$ annual healing draws before reproducing, and at the rates in question that fixes the per-generation recovery probability near one half almost regardless of α. What governs whether adversity persists is how many opportunities for recovery a life affords, rather than the size of the reproductive advantage. Neither discrepancy is a failure to reproduce the analytical result: together they measure what age structure and a graded state contribute, neither of which a mean-field model can represent. We reach the same conclusion below from the endemic levels themselves, by a different route.

Background exposure raises the boundary substantially, to $q^*(\alpha) = 0.032\alpha + 0.156$ at $p = 0.05$ ($R^2 = 0.98$). The positive intercepts in both fits imply a minimum healing requirement that persists even as $\alpha \to 1$, so a population with no reproductive differential at all still requires sustained recovery to suppress adversity.

Transmission dominates the demographic channel

An observation we did not anticipate is how little of the long-run outcome the reproductive differential accounts for. Across the full swept range of α, from 1.05 to 1.50, the mean ACE count rises by between 1.1% and 6.0% depending on the healing rate and on background exposure (Figure 5C–D), while varying q over a comparable range moves it by about 45%. The endemic level is set principally by the intergenerational transmission distribution $\Pr(k' \mid k)$, which is fixed by data, and only secondarily by the demographic mechanism the analytical model is built around. We report this because it bears directly on where intervention effort is best directed, and because it qualifies the analogy above: the demographic amplification term is real and it moves the threshold, but in a realistically parameterised population it is not what determines how much adversity there is.

Because fertility is normalised by the population mean multiplier (Section 2), neither α nor q has a demographic footprint: the growth rate is $-0.0007\,\mathrm{yr}^{-1}$ across every parameter combination we simulate (Figure 4A). This is by construction, and it is the property that lets α mean the same thing in both models. It also means the simulation makes no claim about how adversity affects population size, only about how it is distributed.

A floor that healing cannot reach

Because exposure acts only on children and healing only on adults, sustained childhood exposure sets a floor on population adversity that no adult recovery rate can penetrate. At $\alpha = 1.35$ and a healing probability of 0.60, an implausibly high rate, the mean ACE count still settles at 0.91 when $p = 0.10$ and at 1.37 when $p = 0.20$. Above an exposure rate of roughly 0.1 per childhood year there is therefore no healing rate that brings the population below the threshold at all.

This has a direct reading. Adult recovery and childhood exposure are not two interchangeable levers on the same quantity. Where ongoing exposure is high enough, reducing it is not one option among two but a precondition, and therapeutic provision alone cannot substitute for it however well resourced. The analytical model cannot express this, because it applies both processes to the whole population every generation; it is a consequence of age structure and so appears only in the simulation.

Intervention

To drive the system below endemic persistence, q must exceed $1 - 1/\alpha$, and any intervention that reduces α simultaneously lowers the required healing rate. Interventions combining socioeconomic support with expanded therapeutic access therefore act on both factors in $\alpha(1-q)$. The fitted relationship $q^*(\alpha) = m\alpha + c$ makes the requirement explicit for a given population, and the positive intercept sets a floor below which no reduction in α alone suffices.

We emphasise that $\alpha > 1$ is a descriptive demographic rate. It reflects observed associations between early adversity and accelerated life-history strategies [3, 21] and carries no implication that trauma is adaptive, nor any normative content whatever. The model is an accounting of population-level flows and says nothing about individuals. Its intervention implications run entirely through reducing exposure and supporting recovery.

Limitations

Both models simplify adversity: the analytical model uses a binary state, the simulation a discrete count on $\{0, \ldots, 8\}$, and neither captures continuous severity or multi-domain ACE categorisations. The uniform allocation of probability mass within the reported 2–3 and ≥ 4 transmission categories is an assumption of convenience, and the ceiling at eight binds at high exposure rates, so results at $p \geq 0.1$ are partly a saturation effect.

Mortality is ACE-independent, though evidence links high ACE burden to premature mortality. Whether including it would raise or lower the persistence threshold depends on when the excess mortality falls relative to the reproductive window, and we do not attempt to resolve that here. Transmission is conditioned on maternal exposure only, and no spatial or network structure is included, so the clustering of adversity within families and neighbourhoods is absent; since clustering reduces effective mixing, we expect a structured model to require more healing rather than less.

Finally, the model draws its parameters from four different populations, as set out above. Consistent single-population calibration, and the cross-cultural variation it would expose, remain unexplored.

Conclusion

We have shown that a minimal model of intergenerational adversity admits an explicit persistence threshold, that an age-structured simulation reproduces its dependence on the reproductive differential while requiring substantially more recovery than the mean-field account predicts, and that sustained childhood exposure imposes a floor which adult recovery cannot reach. The framework gives population-level adversity science a formal structure it has so far lacked. Whether real populations sit above or below the threshold is an empirical question, and one we hope to address with cohort data in future work.

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Supplementary Information

.1 Stability of the endemic equilibrium

The main text locates the fixed points of the update map but does not address their stability. We record that here. Write the map as

$$g(x) = \frac{Ax + p}{Bx + 1}, \qquad A = \alpha(1 - q) - p, \qquad B = \alpha - 1.$$

Proposition 1. *Let $\alpha > 1$ and $p, q \in (0, 1)$ with $p + q < 1$. Then g has a unique fixed point x^* in $(0, 1)$, and every orbit starting in $[0, 1]$ converges to it.*

Proof. Since $Bx + 1 \geq 1$ on $[0, 1]$, the map has no pole there. Evaluating at the endpoints gives $g(0) = p$ and $g(1) = \alpha(1 - q)/\alpha = 1 - q$, both in $(0, 1)$, so g maps $[0, 1]$ into itself.

Differentiating,

$$g'(x) = \frac{A - Bp}{(Bx + 1)^2}, \qquad A - Bp = \alpha(1 - q) - p - (\alpha - 1)p = \alpha\big(1 - q - p\big),$$

which is positive when $p + q < 1$, so g is strictly increasing.

Because $Bx + 1 > 0$, the net change $\Delta(x) = g(x) - x$ has the opposite sign to the quadratic $(\alpha - 1)x^2 + (1 + p - \alpha + \alpha q)x - p$. That quadratic has leading coefficient $\alpha - 1 > 0$ and constant term $-p < 0$, so the product of its roots is $-p/(\alpha - 1) < 0$ and exactly one root is positive. Since $\Delta(0) = p > 0$ and $\Delta(1) = -q < 0$, that root lies in $(0, 1)$ and is the unique fixed point there.

Uniqueness and monotonicity together give $\Delta > 0$ on $[0, x^*)$ and $\Delta < 0$ on $(x^*, 1]$. For $x_0 < x^*$ the orbit is therefore increasing, and because g is increasing it remains below x^*; a bounded monotone orbit converges, and its limit is a fixed point, hence x^*. The case $x_0 > x^*$ is symmetric. The boundary cases $x_0 \in \{0, 1\}$ are immediate, since g maps them into the interior. □

The condition $p + q < 1$ is not restrictive: both parameters are annual probabilities well below one half throughout. We verified the proposition numerically on 4,000 randomly drawn parameter sets, in every case recovering a unique interior fixed point with $|g'(x^*)| < 1$ and convergence from both endpoints.

Setting $p = 0$ recovers the threshold of the main text as a stability statement. The trauma-free state $x = 0$ is then a fixed point with $g'(0) = \alpha(1 - q)$, so it is asymptotically stable precisely when $\alpha(1 - q) < 1$. At $\alpha(1 - q) = 1$ it loses stability in a transcritical bifurcation, at which the interior equilibrium $x^* = 1 - \alpha q/(\alpha - 1)$ enters $(0, 1)$. The threshold $q < 1 - \alpha^{-1}$ is thus an exchange of stability between the trauma-free and endemic states, not merely a sign condition.

.2 Relating the two time scales

The parameters p and q do not carry the same units in the two models, and the two thresholds cannot be compared until they do. In the analytical model each acts once per generation on a binary state. In the simulation each acts once per year on a count in $\{0, \ldots, 8\}$, and each is restricted to a phase of life: exposure to childhood, healing to adulthood.

Three separate differences follow. First, the time step: the generation time implied by the fitted schedules, measured as the mean age at childbearing for an unexposed individual, is $T = 27.7$ years. Second, the phase restriction: an individual is subject to healing only between age 18 and reproduction, a window of $D = T - 18 = 9.7$ years, and to exposure only over the 18 years of childhood. Third, the state space: one healing event decrements the ACE count by one, so an individual carrying k ACEs requires k such events to become untraumatised, whereas one recovery event in the analytical model clears the state outright.

Writing q_{yr} and p_{yr} for the annual rates, the number of healing events before reproduction is Binomial(D, q_{yr}), so an individual with k ACEs recovers within one generation with probability

$$q_{\mathrm{gen}}(k) = \Pr\big[\mathrm{Binomial}(D, q_{\mathrm{yr}}) \geq k\big], \qquad q_{\mathrm{gen}}(1) = 1 - (1 - q_{\mathrm{yr}})^D,$$

and correspondingly $p_{\mathrm{gen}} = 1 - (1 - p_{\mathrm{yr}})^{18}$ for acquiring at least one ACE during childhood. The two-state analytical parameters correspond to the $k = 1$ case. For orientation, $q_{\mathrm{yr}} = 0.03$ maps to $q_{\mathrm{gen}} = 0.255$ and $p_{\mathrm{yr}} = 0.02$ to $p_{\mathrm{gen}} = 0.305$: annual rates of a few percent are per-generation probabilities of a few tens of percent, an order of magnitude apart.

We emphasise that this mapping is exact only for the $k = 1$ case in a stationary population. It neglects the dependence of T on k through the tempo effect, and it treats the ACE count as though recovery from $k > 1$ were the only additional obstacle. Any residual discrepancy between the two thresholds after this transformation is therefore not a units artefact: it measures the contribution of age structure and of the graded adversity state, neither of which the mean-field model can represent.

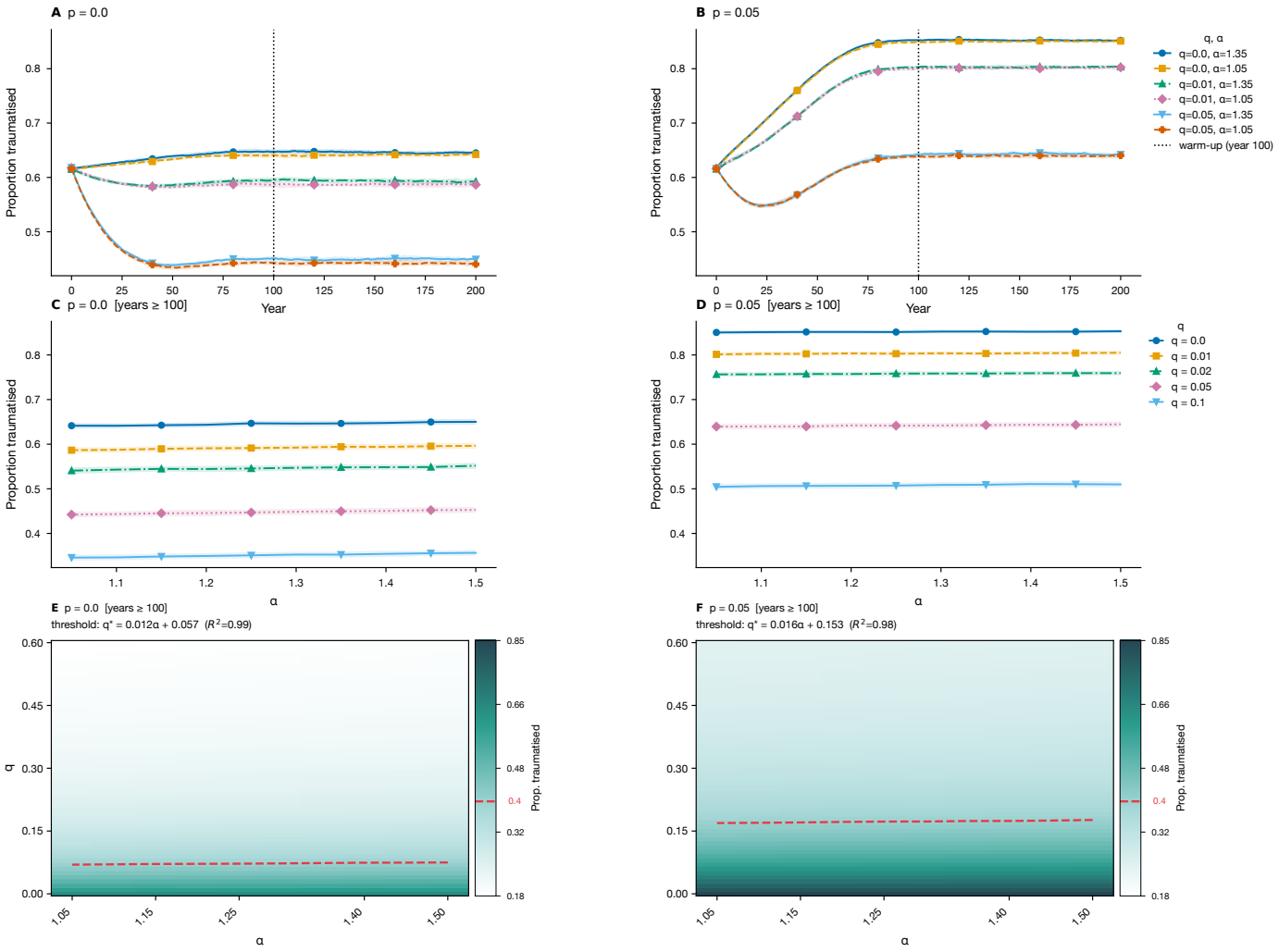


Figure 6: **Simulation outcomes across parameter regimes (proportion traumatised).** Layout mirrors Figure 5. Left column: no background trauma exposure ($p = 0$); right column: positive background exposure ($p = 0.05$). **A**–**B**, Time trajectories of the proportion of individuals with at least one ACE for the same six parameter combinations as Figure 5, shown as median ± IQR. **C**–**D**, Mean ± SD of proportion traumatised as a function of α, one curve per healing probability. **E**–**F**, Heatmaps of the time-averaged proportion traumatised in the (q, α) parameter plane ($\log(1 + \cdot)$ scale). The dashed red contour marks the k-means threshold; the linear fit $q^* = m\alpha + c$ is consistent with the mean-ACE results in Figure 5E–F.

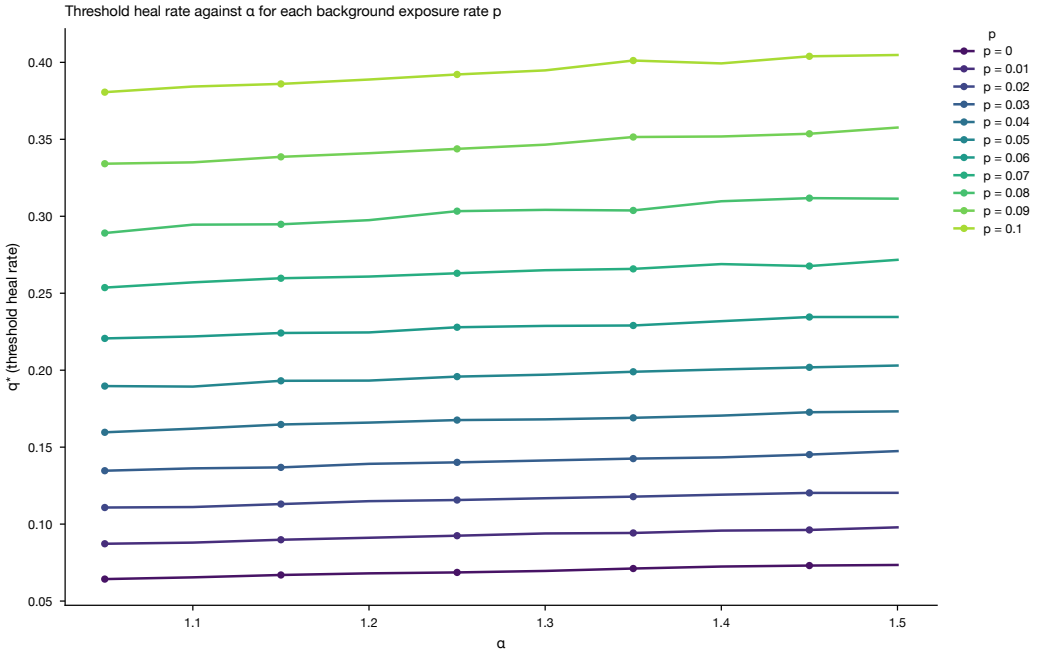


Figure 7: **k-means threshold contours across all values of p.** Each curve shows $q^*(\alpha)$, the healing probability at which the mean ACE surface crosses the threshold, for a single value of the background exposure probability p (identified by colour; see legend). A single threshold is used throughout, computed by k-means ($k = 2$) on the $\log(1 + \cdot)$-transformed surface pooled over $p = 0$ and $p = 0.05$, and crossings are located by linear interpolation along the q-axis. The contours are therefore directly comparable in level with Figure 5E and F, whose red contours the $p = 0$ and $p = 0.05$ curves reproduce. Higher p shifts the threshold upward, from $q^* \approx 0.064$ at $p = 0$ to $q^* \approx 0.38$ at $p = 0.1$, both evaluated at $\alpha = 1.05$. At $p = 0.2$ the surface lies wholly above the threshold, so no healing rate in the swept range crosses it and no contour is drawn.

.3 Simulation outcomes: proportion-traumatised metric

.4 k-means threshold across all values of p

.5 k-means threshold procedure

To identify the boundary between parameter regimes that sustain high and low levels of adversity, we applied k-means clustering with $k = 2$ [10] to the log-transformed mean ACE surface. Let $\bar{k}(q, \alpha)$ denote the mean ACE count averaged across all years and simulation replicates at healing probability q and reproductive multiplier α, and let $\ell(q, \alpha) = \log\big(1+\bar{k}(q, \alpha)\big)$ be its $\log(1+\cdot)$ transform, applied to compress the dynamic range of the surface and to handle zero values stably. k-means partitions the observed values of ℓ into two clusters by iteratively assigning each value to its nearest cluster centre and updating centres to cluster means until convergence. We initialise with k-means++ [1], keep the best of ten restarts, and fix the random seed so that the threshold is reproducible. The scalar threshold ℓ^* is the midpoint between the two converged cluster centres, and requires no user-specified cutoff. We compute ℓ^* once, on the values of ℓ pooled over $p = 0$ and $p = 0.05$, and apply the same ℓ^* at every value of p, so that the threshold heal rates reported in Figure 5 and Figure 7 are on one scale and can be compared directly. The threshold defines a level set $\{(q, \alpha) : \ell(q, \alpha) = \ell^*\}$ in the original parameter space, which we locate for each value of α by linear interpolation along the q-axis to obtain crossing points $q^*(\alpha)$. Although the threshold is identified in log-space, the resulting boundary is expressed in the untransformed coordinates (q, α), and is well described by a linear relationship

$$q^*(\alpha) = m\alpha + c,$$

recovered by ordinary least squares regression on the interpolated crossing points (fits shown in panel titles; $R^2 > 0.95$ in all cases). The boundary is therefore close to linear in α over the swept range, with a positive intercept. It does not follow the analytical threshold $q = 1-\alpha^{-1}$ derived in Section 2: once both are expressed per generation (Supplementary Information .2) the simulated boundary is higher and markedly flatter, as we discuss in Section 4. Figure 7 shows the resulting threshold contours for every value of p in the data whose surface crosses ℓ^*.